

An XR-based Approach to Safe Human-Robot Collaboration

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ABSTRACT

It is crucial to prevent safety accidents for human-robot collaboration. This study proposes an extended reality (XR) approach to safe HRC by calculating the minimum distance between the human operator and robot in real time. The proposed approach scans the real environment with multiple sensors and constructs its virtual space in XR. The virtual robot is synchronized with the real robot by matching the point cloud of real environment with that of the virtual robot. This system can effectively provide task assistance and safety information to the user wearing an XR device.

Keywords: Human-robot collaboration, mixed/extended reality (MR/XR), safety distance.

Index Terms: H.5.1 [Information Interfaces and Presentation]: Multimedia Information Systems—XR-based safety information; J.7 [Computer Applications]: Computers in Other Systems—Virtual-Physics synchronization; I.3.8 [Computer Graphics]: Applications—Extended Reality

1 INTRODUCTION

Since the 4th industrial revolution was introduced, the interest in human-robot collaboration (HRC) has increased significantly. Humans and robots can use each other's strengths in most HRC situations. Humans and robots can complement each other's weaknesses and work together as collaborative partners. Humans have excellent ability to adapt and cope with various conditions, whereas robots can repeatedly perform predetermined movements accurately and quickly [1]. Therefore, humans can focus on more creative tasks while robots perform repetitive tasks. However, it is necessary to recognize the user's situation and provide appropriate feedback to provide user-oriented and safe HRC in uncertain and dynamic scenarios. Thus, a new method should be designed and implemented through the combination of the Internet of Things (IoT), cyber-physical systems (CPS), extended reality (XR), and artificial intelligence (AI) [2].

This study proposes a safe human-robot collaboration (HRC) method by synchronizing a real robot and its corresponding virtual robot in an XR environment, providing more safety-aware and user-centric task assistance. To construct an XR-based HRC

environment, the proposed approach scans the real environment with multiple sensors and synchronizes the real robot with the virtual robot. Synchronization is conducted by matching the scanned point cloud of the real robot with that of the virtual robot in XR.

2 XR-BASED APPROACH TO HRC

An accident may occur when the operator cannot predict the robot's movement and enters the robot's working area. To secure human safety during HRC in the XR-based environment, it is crucial to measure the safety distance between the worker and robot in real-time. Besides, it is necessary to assist the worker wearing XR glasses to conduct required tasks readily and effectively. In particular, the XR-based visual information for HRC should be augmented on the physical robot seamlessly.

2.1 Synchronization of Virtual and Physical Spaces

The proposed HRC approach in XR is shown in Figure 1. When the operator collaborates with the robot, the RGB image and point cloud data are acquired through the RGBD sensors installed in the surrounding real environment. Two Microsoft Azure Kinects are used to scan the environment [3]. Deep learning-based segmentation and 3D human pose estimation are conducted to synchronize the physical robot with the XR-based HRC environment and calculate the safety distance between the operator and robot, respectively. Segmentation finds the area of a specific object in the XR environment. In this study, Detectron2 [4] is used to conduct segmentation from RGB images captured from the sensor.

ICP matching is performed to synchronize the segmented point cloud of the real robot with that of the virtual robot in XR. ICP matching serves 3D matching between different point cloud groups [5]. In addition, the 3D human skeleton information is extracted from the RGB-D data by applying the 3D human pose estimation method. The skeleton information is used to calculate the minimum distance between the robot and the human. The minimum distance is used to stop or slow down the robot if they are too close to each other. In addition, the minimum distance is communicated to the operator through the XR glasses, such as HoloLens 2 [6].

2.2 Safety Distance Calculation

Since it is crucial to maintain safety during HRC, it is essential to calculate the distance between the robot and human in real-time. To this end, the proposed approach calculates the minimum distance between the virtual robot's link and the human skeleton in XR instead of measuring the distance between their point clouds. The distance is obtained from the 3D offset of the line segments of the robot and the human skeleton [2]. Details are described in Choi et al. (2022) [2].

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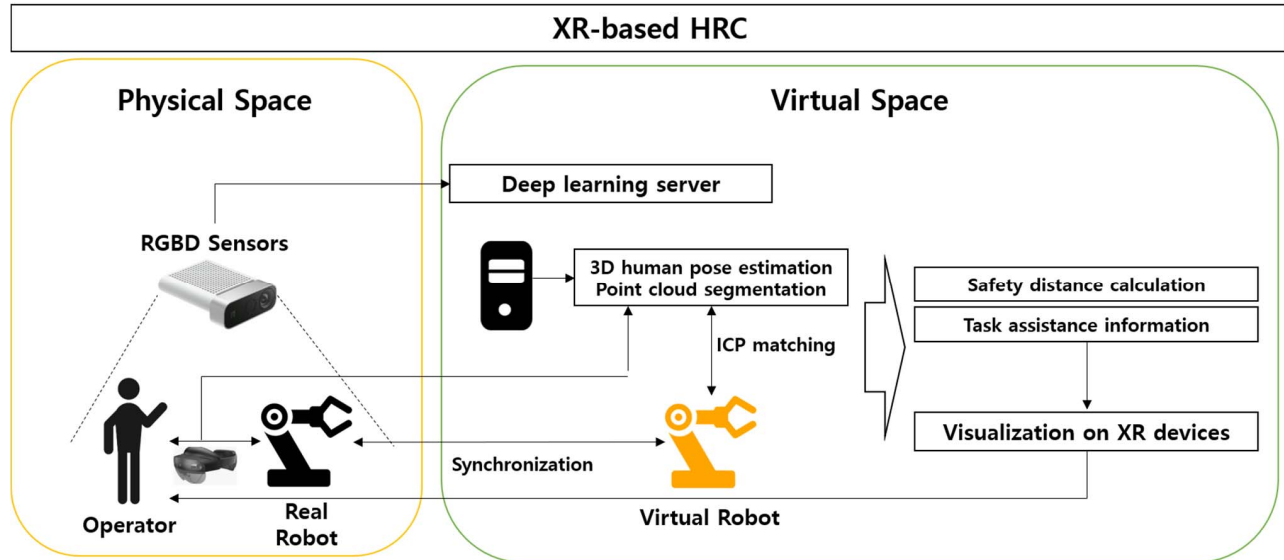


Figure 1. Overview of XR-based approach to human-robot collaboration.

2.3 System Implementation

Figure 2 shows the implementation results of the proposed approach, which can provide various task assistance and safety information to the user during XR-based HRC. Figure 2 (a) shows the visualization of the human skeleton in XR when the operator interacts with the real robot. Figure 2(b) depicts the synchronized visualization of the physical and virtual robots and virtual robot simulation. Figure 2(c) shows the minimum distance calculation in XR. Through the proposed XR-based HRC, the human operator can conduct required tasks more safely when a safety warning is sent to the user when the minimum distance is within a given limit. Furthermore, the user can conduct given tasks more quickly and effectively by visualizing the robot's movement in advance through virtual robot simulation and acquiring task-related information augmented onto real objects.

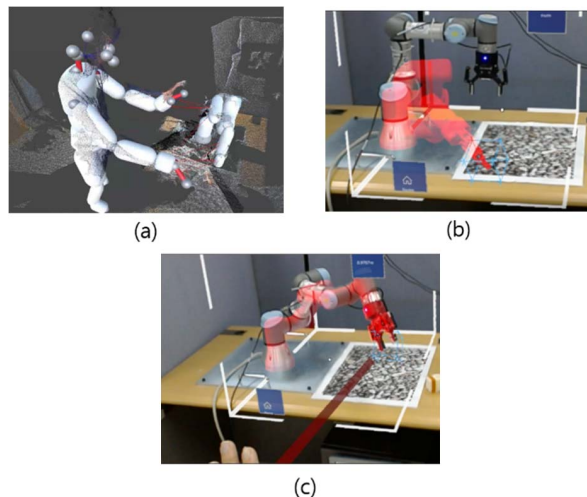


Figure 2. Implementation results: (a) synchronized human operator and virtual robot in XR, (b) virtual robot simulation synchronized with the real robot, (c) safety distance calculation and visualization.

3 CONCLUSION

This study proposed an XR-based HRC application to assist the human operator in conducting HRC tasks more effectively and safely. The proposed approach measures the safety distance between the human operator and robot in real-time by tracking the human skeleton and synchronized robot. The synchronization among the virtual robot, human operator, and the real environment is conducted by deep learning-based segmentation and ICP matching. We will extend 3D human pose estimation to classify human action and apply the proposed approach to various industrial applications for future research work.

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